

**PROBABILITY & STATISTICS (MT-331)**  
**Department :SE(SOFTWARE) BATCH 2026**

**HANDOUT#1**

1. Find the mean, median and mode of each set of data & write Comments on the shape of the Distribution:

A) 24,90,10,78,50,35,90,33,46

B) 5,9,1,6,4,8,3,5,6,9,7,4,8,5,5,1

C) 100,98,155,110,99,123,140,145

2- Choose the correct stem and leaf plot which represent the following data

16 10 11 30 35 33 35 44 47

a) 

|   |         |
|---|---------|
| 1 | 0 1 6   |
| 2 | 0       |
| 3 | 0 3 5 5 |
| 4 | 4 7     |

c) 

|   |         |
|---|---------|
| 1 | 0 1 6   |
| 2 |         |
| 3 | 0 3 5 5 |
| 4 | 4 7     |

b) 

|   |         |
|---|---------|
| 1 | 0 1 6   |
| 3 | 0 3 5 5 |
| 4 | 4 7     |

d) 

|   |       |
|---|-------|
| 1 | 0 1 6 |
| 3 | 0 3 5 |
| 4 | 4 7   |

Q3-(a) Draw frequency distribution table with the help of graph?

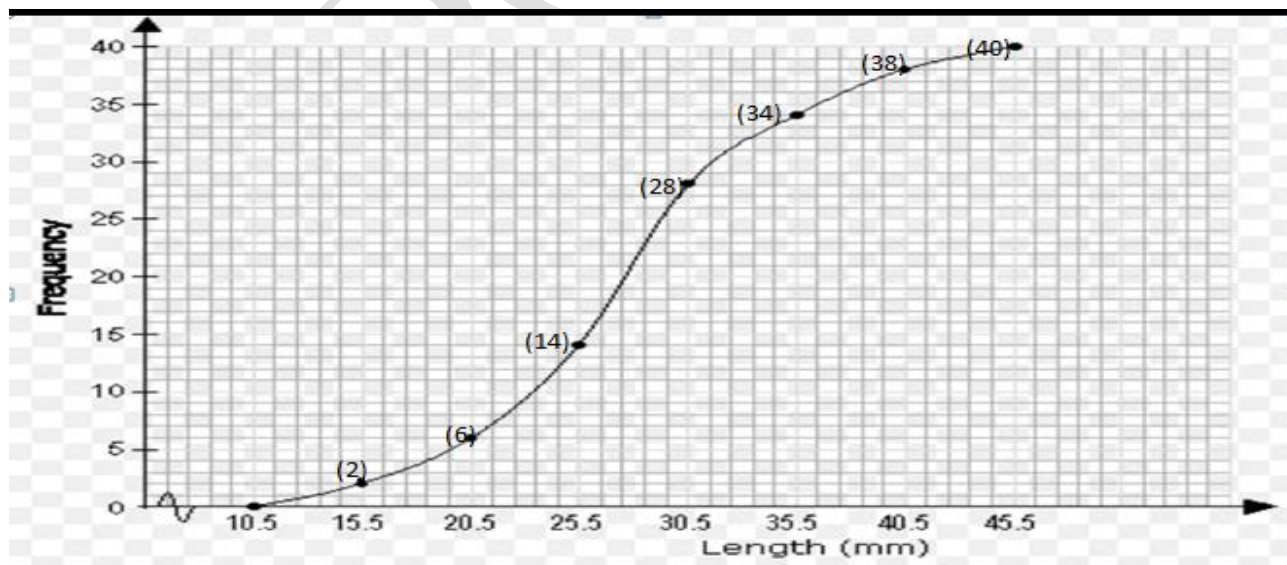
(b) Calculate Q1 Q2 & Q3 & write comments with the help of Quartiles?

(c) Verify result with the help of Graph?

(d) Calculate mode?

(e) Calculate inter quartile range and quartile deviation. Also calculate 9<sup>th</sup> Decile?

(f) Calculate variance?

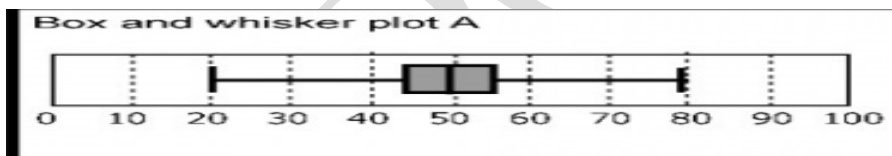


Over a period of 60 days the percentage relative humidity in a vegetable storage building was measured. Mean daily values were recorded as shown below:

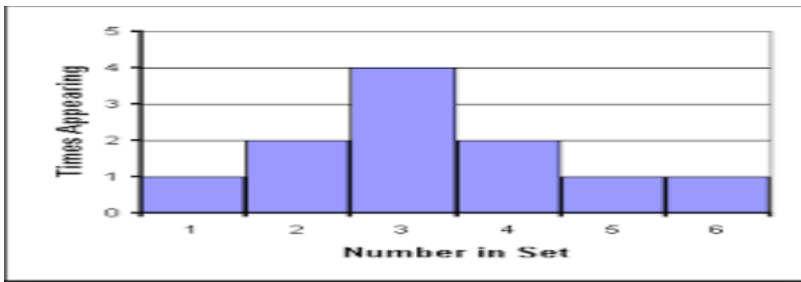
|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 60 | 63 | 64 | 71 | 67 | 73 | 79 | 80 | 83 | 81 |
| 86 | 90 | 96 | 98 | 98 | 99 | 89 | 80 | 77 | 78 |
| 71 | 79 | 74 | 84 | 85 | 82 | 90 | 78 | 79 | 79 |
| 78 | 80 | 82 | 83 | 86 | 81 | 80 | 76 | 66 | 74 |
| 81 | 86 | 84 | 72 | 79 | 72 | 84 | 79 | 76 | 79 |
| 74 | 66 | 84 | 78 | 91 | 81 | 64 | 76 | 78 | 82 |

- Make a stem-and-leaf display with at least five stems for these data. Show the leaves sorted in order of increasing magnitude on each stem.
- Make a frequency table for the data, with a maximum bound of 100.5% relative humidity (since no relative humidity can be more than 100%). Use Sturges' rule to approximate the number of classes.
- Draw a frequency histogram for these data.
- Draw a relative cumulative frequency diagram.
- Find the median, lower quartile, and upper quartile.
- Find the arithmetic mean of these data.
- Find the mode of these data from the grouped frequency distribution.
- Draw a box plot for these data.
- Calculate inter quartile range?
- Calculate variance & coefficient of variance?
- Calculate skewness & kurtosis & write comments on the shape of the distribution?

- Q5 (a) with the help of box plot calculate five number summary?  
 (b) Calculate inter quartile range and quartile deviation?



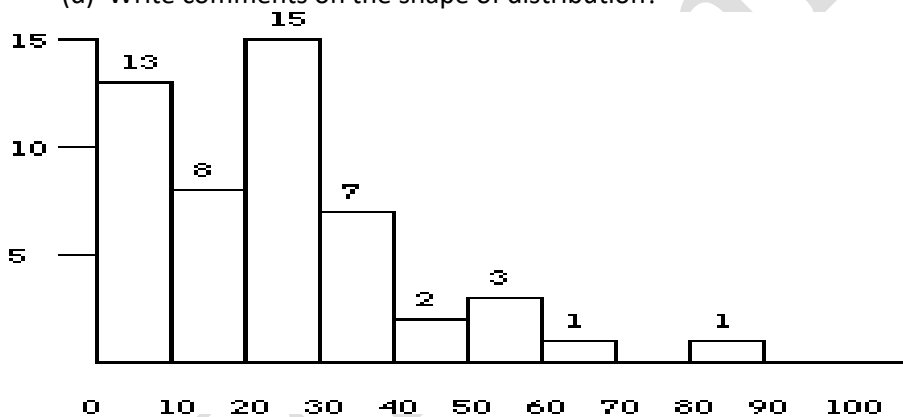
- Q6 (a) draw frequency distribution table with the help of graph?  
 (b) Calculate skewness & kurtosis with the help of moment about mean?  
 (c) Write comments on the shape of distribution?  
 (d) Calculate coefficient of variation?  
 (e) Calculate 99% range of the given data?



- Q7 (a) calculate inter quartile range?  
 (b) Calculate median?

| stem | leaf    |
|------|---------|
| 10   | 0       |
| 9    | 4 5 5 8 |
| 8    | 5 5 5   |
| 7    | 4 7     |

- Q8 (a) With the help of histogram draw frequency distribution table  
 (b) Calculate mean median & mode?  
 (c) Calculate variance & 68% range of the given data?  
 (d) Write comments on the shape of distribution?



Q9-Thirty AA batteries were tested to determine how long they would last. The results, to the nearest minute, were recorded as follows:

423, 369, 387, 411, 393, 394, 371, 377, 389, 409, 392, 408, 431

- a) Calculate Five number summary?  
 b) Write comments on the shape of the distribution with the help of quartile?  
 c) Calculate moment about origin & with the help of moment about origin calculate skewness & kurtosis & write comments?

Q10-A survey was taken on Maple Avenue. In each of 20 homes, people were asked how many cars were registered to their households. The results were recorded as follows: Calculate Skewness & Kurtosis With the help of moment about mean & moment about origin & compare both results.

1, 2, 8, 0, 3, 4, 0, 8, 1, 1, 2, 2, 3, 2, 3, 2, 1, 8, 10

### Example 8 ▶ Sketching Box-and-Whisker Plots

Sketch a box-and-whisker plot for each set.

- 27, 28, 30, 42, 45, 50, 50, 61, 62, 64, 66
- 82, 82, 83, 85, 87, 89, 90, 94, 95, 95, 96, 98, 99
- 11, 13, 13, 15, 17, 18, 20, 24, 24, 27

#### Solution

- This set has 11 numbers. The median is 50 (the sixth number). The lower quartile is 30 (the median of the first five numbers). The upper quartile is 62 (the median of the last five numbers). See Figure A.7.



FIGURE A.7

- This set has 13 numbers. The median is 90 (the seventh number). The lower quartile is 84 (the median of the first six numbers). The upper quartile is 95.5 (the median of the last six numbers). See Figure A.8.

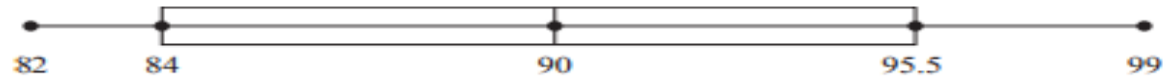


FIGURE A.8

- This set has 10 numbers. The median is 17.5 (the average of the fifth and sixth numbers). The lower quartile is 13 (the median of the first five numbers). The upper quartile is 24 (the median of the last five numbers). See Figure A.9.



FIGURE A.9

- The following table gives the number of weeks needed to find a job for 20 older workers that lost their jobs as a result of corporation downsizing. Find Five number summary (min., Q1, median, Q3, max.) of the data and Interpret your answer

|    |    |    |    |    |
|----|----|----|----|----|
| 13 | 13 | 17 | 7  | 22 |
| 22 | 26 | 17 | 13 | 14 |
| 16 | 7  | 6  | 18 | 20 |
| 10 | 17 | 11 | 10 | 15 |

- Find the mean deviation of the set (a) 3, 7, 9, 5 and (b) 2.4, 1.6, 3.8, 4.1, 3.4.

For the set 8, 10, 9, 12, 4, 8, 2, find the mean deviation from the mean and five number summary?

14 Goals scored by two team A & B in a football season were as follows

| No of goals scored in Match ( $x_i$ ) |   | 0  | 1 | 2 | 3 |
|---------------------------------------|---|----|---|---|---|
| Number of Matches                     | A | 27 | 9 | 8 | 5 |
|                                       | B | 17 | 9 | 6 | 5 |

Find out mean and standard deviation & tell which team may be considered more consistent

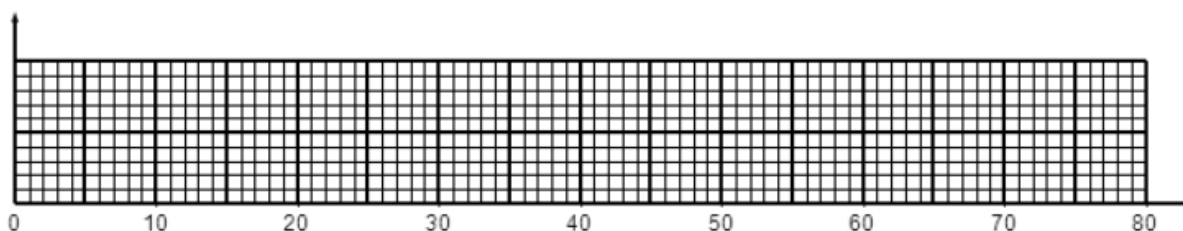
15 The distribution of SAT scores for a group of high school students has a first quartile score equal to 825 and a third quartile score equal to 1125. Calculate the quartile deviation for the distribution of SAT scores for this group of high school students.

### Example

The number of passengers in a 72-seater aeroplane was noted on each of its 20 flights in one day.

48, 54, 72, 21, 37, 45, 72, 41, 59, 63, 54, 59, 68, 72, 70, 64, 41, 32, 24, 31

- Work out the three quartiles for this data.
- Draw a box plot for this data.



### Stem and Leaf Plots/Diagrams

A **stem and leaf** plot is a **frequency diagram** in which the raw data is displayed together with its frequency. The data is then placed in suitable groups.

**Example.** The following data gives the marks out of 60 for a maths test. Place the data in a stem and leaf diagram.

28, 38, 42, 5, 13, 23, 14, 38, 56, 20, 32, 47, 58, 3, 18, 19, 42, 48, 29, 14, 24, 28, 50, 31, 35

